

Task 3 - Data Analysis & Insight Extraction

Step 1 - Describe

Prompt: Describe the cleaned sales dataset and its fields. Key output: 20 orders across North, South, East and West, with four products and sales from March 1-15.

Step 2 - Summarize

Prompt: Summarize total revenue and main sales measures. Key output: total recorded/cleaned revenue is 15,169.57; South contributes 7,679.85.

Step 3 - Rank

Prompt: Rank regions and products by revenue. Key output: regions: South 7,679.85, West 3,639.89, North 3,179.83, East 670.00. Products: Widget A 6,390.00, Gadget X 3,869.57, Service Pack 2,750.00, Widget B 2,160.00.

Step 4 - Trend

Prompt: Show the daily revenue trend and identify peaks. Key output: revenue rises sharply on March 15 mainly because order 1019 contributes 4,455.00.

Step 5 - Anomaly

Prompt: Find anomalies and data points that may distort results. Key output: order 1019 qty=99 is the main outlier; pending/missing rows need caution.

Step 6 - Insight

Prompt: Give five actionable business insights using only the cleaned data. Key output: South and Widget A lead revenue, East is weak, the March 15 peak is outlier-driven, and data quality controls are needed.

Developer analysis script request

Write a pandas + matplotlib script using sales_clean.csv. Create one bar chart for total revenue by region, one line chart for daily revenue, and export summary statistics to analysis_report.xlsx. The script must convert date/revenue safely and run end-to-end.

#	Insight in my words	Supporting data	Recommended action
1	South is the strongest region.	Revenue 7,679.85, about half of total.	Study what is working in South and reuse it elsewhere.
2	Widget A is the top revenue product.	Widget A revenue is 6,390.00.	Protect supply and promote it, but verify the large order.
3	East is much weaker than other regions.	East revenue is only 670.00.	Review East pipeline, pending orders and sales activity.
4	March 15 performance is heavily affected by one order.	Order 1019 revenue is 4,455.00 with qty 99.	Verify order 1019 before using it for forecasting.
5	Data quality can change conclusions.	Missing qty/revenue and inconsistent formats exist.	Add validation before every weekly report.